

Gas Kinetics

Some fundamental numbers to know are:

$A \Rightarrow$ Avogadro's # $\left[\frac{\text{\# molecules}}{\text{kmol}} \right] = 6.022 \times 10^{26}$

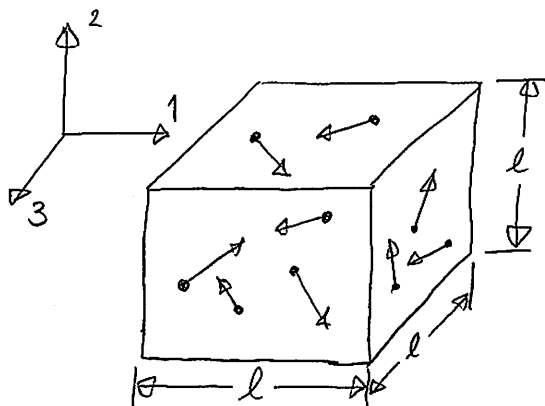
$R \Rightarrow$ Universal gas constant $\left[\frac{\text{J}}{\text{kmol} \cdot \text{K}} \right] = 8314.3$

$k_B \Rightarrow$ Boltzmann constant $\left[\frac{\text{J}}{\text{molecule} \cdot \text{K}} \right] = 1.38 \times 10^{-23} \Rightarrow$ Relates individual molecule energy to temperature

$M \Rightarrow$ Molecular weight $\left[\frac{\text{kg}}{\text{kmol}} \right]$

$$k_B = \frac{R}{A}$$

We can now develop a model to give meaning for pressure in a rarified gas (non-interacting gas molecules)



$$V = l^3$$

N molecules in our volume
 $l < \lambda_i$ (Rarified gas)
 $m \Rightarrow$ mass per particle

We have a particle traveling at C_{ii}

The change in momentum from a collision with the wall normal to 1 is $2mC_{ii} = \Delta p$

Aside: i 'th particle with velocity component in 1 direction

$$\Delta p = p_i - p_f = [C_{ii} - (-C_{ii})]m = 2mC_{ii}$$

Relating this to force: $F = \frac{\Delta p}{\Delta t}$, or as you know it $F = \frac{\partial p}{\partial t}$

Δt can be related to the frequency of particle collisions, f , with the wall.

$$f = \frac{C_{ii}}{2l} \Rightarrow \Delta t = \frac{1}{f} = \frac{2l}{C_{ii}}$$

$$F_i = \frac{2mC_{ii}}{\frac{2\ell}{C_{ii}}} = \frac{mC_{ii}^2}{\ell}$$

$$F_1 = \sum F_i = \frac{m}{\ell} \sum_{i=1}^N C_{ii}^2$$

$$P_1 = \frac{F_1}{\ell^2} = \frac{m}{\ell^3} \sum_{i=1}^N C_{ii}^2$$

(Pressure)

So for our three surfaces:

$$P_1 = \frac{m}{\ell^3} \sum_{i=1}^N C_{ii}^2; \quad P_2 = \frac{m}{\ell^3} \sum_{i=1}^N C_{2i}^2; \quad P_3 = \frac{m}{\ell^3} \sum_{i=1}^N C_{3i}^2$$

The total pressure in our chamber can be found by taking the average of our 3 directional pressures.

$$P = \frac{P_1 + P_2 + P_3}{3} = \frac{m}{\ell^3} \frac{1}{3} \sum_{i=1}^N \underbrace{C_{ii}^2 + C_{2i}^2 + C_{3i}^2}_{C_i^2}$$

$$P = \frac{m}{\ell^3} \frac{1}{3} \sum_{i=1}^N C_i^2$$

From statistical thermodynamics, we know that:

$$\underbrace{\frac{1}{2} m \bar{C}^2}_{\text{average molecule KE}} = \underbrace{\frac{3}{2} k_B T}_{\frac{1}{2} k_B T \text{ for every translational degree of freedom}}$$

$$\text{We also know that: } m = \frac{M}{A} \left[\frac{\text{kg}}{\text{molecule}} \right]$$

$$\frac{1}{2} m \bar{C}^2 = \frac{3}{2} k_B T = \frac{1}{2} \frac{M}{A} \bar{C}^2$$

$$M \bar{C}^2 = 3 k_B A T \Rightarrow \text{we know that } k_B A = R$$

$$\boxed{M \bar{C}^2 = 3RT} \quad (1)$$

Another more rigorous way to prove:

$$F_1 = \frac{m}{\ell} \sum_{i=1}^N C_{ii}^2$$

$$= \frac{M}{A\ell} \sum_{i=1}^N C_{ii}^2$$

$$= \frac{MN_{\text{mol}}}{\ell} \left\{ \frac{\sum_{i=1}^N C_{ii}^2}{N} \right\} \bar{C}_1^2$$

$$F_1 = \frac{MN_{\text{mol}}}{\ell} \bar{C}_1^2, \text{ we know that}$$

Due to random motion: $\bar{C}_1^2 = \frac{1}{3} \bar{C}^2$

Looking back at our pressure term

$$\begin{aligned}
 p &= \frac{m}{\ell^3} \cdot \frac{\sum c_i^2}{3} \\
 &= \frac{M}{A \ell^3} \cdot \frac{\sum c_i^2}{3} \Rightarrow A = \frac{N}{N_{\text{mol}}} \left[\frac{\# \text{ molecules}}{\# \text{ kmol}} \right] \\
 &= \frac{N_{\text{mol}}/M}{N \ell^3} \frac{\sum c_i^2}{3} \Rightarrow \text{Since } \frac{\sum c_i^2}{N} = \overline{c^2} \text{ (average molecule speed)} \\
 &= \underbrace{\frac{N_{\text{mol}}/M}{\ell^3}}_p \cdot \frac{1}{3} \overline{c^2} = \frac{1}{3} p \overline{c^2} \Rightarrow \boxed{p = \frac{1}{3} p \overline{c^2}} \quad (2)
 \end{aligned}$$

Now we can equate ① and ② & isolate for $\overline{c^2}$

$$\overline{c^2} = \frac{3p}{p} = \frac{3RT}{M}$$

$$p = \frac{p}{M} RT, \text{ note } p \Rightarrow \left[\frac{\text{kg}}{\text{m}^3} \right]; M \Rightarrow \left[\frac{\text{kg}}{\text{kmol}} \right]$$

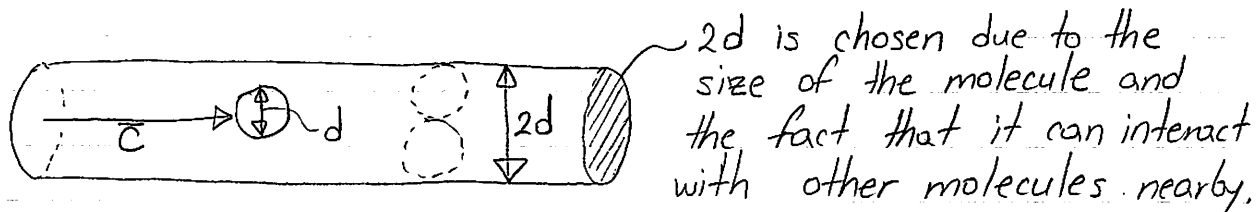
$$\boxed{p = p_m RT} \Rightarrow \text{Ideal Gas Law} \Rightarrow p_m = \text{molecular density} \left[\frac{\text{kgmol}}{\text{m}^3} \right]$$

or

$$\boxed{pV = N_{\text{mol}} RT}$$

$R = \text{universal gas const.}$
 $p = \text{pressure.}$

Now if we look at the microscopic picture of the molecule



We could have chosen $3d$, however there is a chance that the molecule doesn't interact at all then with molecules within the swept path.

$$\frac{\text{Swept Volume}}{\text{time}} = \pi d^2 \cdot c$$

$$\frac{\# \text{ collisions}}{\text{time}} = \pi d^2 n \cdot c, \quad n \Rightarrow \left[\frac{\# \text{ molecules}}{\text{volume}} \right]$$

$$\lambda = \text{mean free path} = \frac{\text{distance traveled/time}}{\# \text{ collisions/time}}$$

$$\lambda = \frac{\cancel{c}}{\pi d^2 n \cancel{c}} = \frac{1}{\pi d^2 n}$$

$$\boxed{\lambda = \frac{1}{\pi d^2 n}} \Rightarrow \text{Mean free path}$$

We can work with this a bit further to get it into a more usefull form: (using our ideal gas law)

$$PV = N_{\text{mol}} RT \Rightarrow A = \frac{N}{N_{\text{mol}}} \Rightarrow N_{\text{mol}} = \frac{N}{A}$$

$$PV = \frac{N}{A} RT = \underbrace{\left(\frac{R}{A} \right)}_{k_B} NT$$

$$PV = k_B NT$$

$$\frac{N}{V} = n = \frac{P}{k_B T} \Rightarrow \text{Back substituting}$$

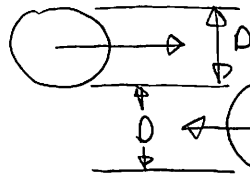
$$\boxed{\lambda = \frac{k_B T}{\pi d^2 P}}$$

If we look at our overall statistical distribution

$$\boxed{\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}} \Rightarrow \begin{matrix} \text{As } T \uparrow, \lambda \uparrow \\ P \downarrow, \lambda \uparrow \end{matrix}$$

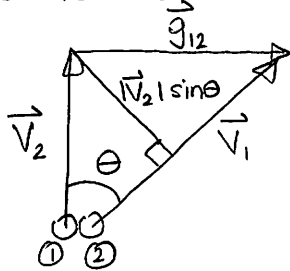
$\hookrightarrow \sqrt{2}$ comes from molecules having a velocity relative to one another.

Extra Derivation (Where the $\sqrt{2}$ in λ comes from)



This molecule is not stationary, it will have a speed as well.

We need to know the mean relative speed between molecules: (consider two arbitrary molecules)



$$|g_{12}| = [|V_2|^2 \sin^2 \theta + (|V_1| - |V_2| \cos \theta)^2]^{1/2}$$

Rearranging, we obtain

$$|g_{12}|^2 = |V_2|^2 + |V_1|^2 - 2|V_1||V_2| \cos \theta$$

Taking $\bar{g}$ to be the root mean square value of $|g_{12}|$, we have

$$\bar{g} = \langle |g_{12}|^2 \rangle^{1/2} = \langle |V_2|^2 + |V_1|^2 - 2|V_1||V_2| \cos \theta \rangle^{1/2}$$

In our model, we assumed $|V_1| = |V_2| = C$

$$\bar{g} = \langle |g_{12}|^2 \rangle^{1/2} = \langle C^2 + C^2 - 2C^2 \cos \theta \rangle^{1/2} = 2C^2 (1 - \langle \cos \theta \rangle)^{1/2}$$

If we average from $\theta = 0$ to $\theta = 2\pi \Rightarrow \langle \cos \theta \rangle = \frac{1}{2\pi} \int_0^{2\pi} \cos \theta d\theta$

$$\therefore \boxed{\bar{g} = (2C^2)^{1/2} = \sqrt{2} C} \Rightarrow \text{Use this in \# of collisions/time equation.}$$

Example #1 / If we look at air at 273K

$$M = 28.9 \text{ kg/kmol}$$

$$\rho_{\text{air}} = 1.288 \text{ kg/m}^3$$

$$\rho_{\text{liquid, air}} = 0.86 \times 10^3 \text{ kg/m}^3$$

$$A = 6.022 \times 10^{26} \text{ molecules/kmol}$$

$$R = 8314.3 \text{ J/kmol} \cdot \text{K}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

First we can calculate the mass per molecule (m)

$$m = \frac{M}{A} = 4.8 \times 10^{-26} \text{ kg/molecule}$$

$$n = \frac{\rho}{m} = 2.684 \times 10^{25} \text{ molecules/m}^3$$

$$\rho = 1.01325 \times 10^5 \text{ N/m}^2 \Rightarrow \text{Aside:}$$

Assuming that the volume of the air molecule can be approximated as d^3

$$\frac{1}{d^3} = n_{\text{liq}} = \frac{\rho_{\text{liq}}}{m}$$

$$d = \left(\frac{m}{\rho_{\text{liq}}} \right)^{1/3} = 3.83 \times 10^{-10} \text{ m}$$

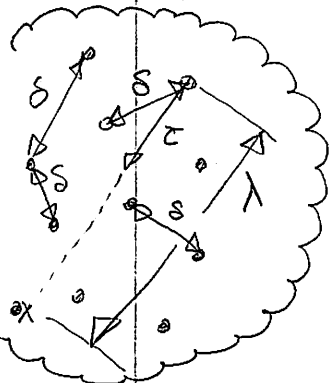
$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n} = 5.717 \times 10^{-8} \text{ m} = \boxed{57.2 \text{ nm} = \lambda}$$

Distance between collision

$$\delta = \frac{1}{n^{1/3}} = 3.35 \times 10^{-9} \text{ m} = \boxed{3.35 \text{ nm}} \text{ (molecular spacing)}$$

Not same as λ .

Picture:



$$\bar{c}^2 = \frac{3P}{\rho} = \frac{3(101325 \text{ Pa})}{(1.288 \text{ kg/m}^3)}$$

$$\bar{c} \approx \sqrt{\bar{c}^2} = 486 \text{ m/s}$$

$$\approx 1750 \text{ km/hr}$$

$\Downarrow$
2.5x faster than a commercial jet plane.

How do we feel this:
With static pressure

$$P_s = \frac{1}{2} \rho_{\text{air}} \bar{c}^2 \approx 100000 \text{ Pa}$$

Works!

Note, we showed earlier that $T = \frac{M \bar{c}^2}{3 k_B A}$, so why doesn't your temperature increase when you're flying in a plane?

Maxwellian Velocity Distribution

$$\phi(c_i) = \frac{1}{\sqrt{\pi}} \left(\frac{m}{2k_B T} \right)^{1/2} e^{-\left(\frac{mc_i^2}{2k_B T} \right)}$$

$\psi(E) = \frac{1}{Ae^{E/k_B T}} = \text{probability a particle has energy } E$
 $\rightarrow \text{Constant}$
 $S = k_B \ln W$
 $0 < \phi(c_i) < 1$

$\rightarrow$ probability of a particle or molecule having a velocity c going in the i th direction (1D)

Like any other distribution:

$$\int_{-\infty}^{\infty} \phi(c_1) dc_1 = \int_{-\infty}^{\infty} \phi(c_2) dc_2 = \int_{-\infty}^{\infty} \phi(c_3) dc_3 = 1$$

We know that: $\frac{2}{\sqrt{\pi}} \int_0^x e^{-x^2} dx = \text{erf}(x)$, $\text{erf}(\infty) = 1$, $\text{erf}(0) = 0$

$$\int_0^{\infty} e^{-c^2} dc = \frac{\sqrt{\pi}}{2}$$

Now let's take a closer look:

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \left(\frac{m}{2k_B T} \right)^{1/2} e^{-\left(\frac{mc_i^2}{2k_B T} \right)} dc_i \Rightarrow \text{Let } c \left(\frac{m}{2k_B T} \right)^{1/2} = \hat{c}$$

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\hat{c}_i^2} d\hat{c}_i = \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-\hat{c}_i^2} d\hat{c}_i = 1 \quad dc = \frac{1}{\left(\frac{m}{2k_B T} \right)^{1/2}} d\hat{c}$$

$$\int_0^{\infty} e^{-\hat{c}_i^2} d\hat{c} = \frac{\sqrt{\pi}}{2} = I_0$$

$\rightarrow$ Multiplied by 2 and changed lower bound to 0 since distribution is symmetric and centered about 0.

Now if we look at higher power integrals:

$$I_m = \int_0^{\infty} c^n e^{-c^2} dc = \int_0^{\infty} \underbrace{c^{n-1}}_u \cdot \underbrace{c e^{-c^2}}_{dv} dc$$

Using integration by parts $\int u dv = uv - \int v du$

$$\int dv = \int c e^{-c^2} dc \Rightarrow \text{Let } u = c^2 \Rightarrow \int \cancel{c} e^{-u} \frac{du}{2\cancel{c}} = \frac{1}{2} \int e^{-u} du$$

$\Rightarrow v = -\frac{1}{2} e^{-c^2}$

$$\frac{dv}{dc} = -\frac{1}{2} (-2c) e^{-c^2} = c e^{-c^2} \Rightarrow dv = c e^{-c^2} dc \Rightarrow \text{WORKS!}$$

$$u = c^{n-1}$$

$$du = (n-1) c^{n-2} dc$$

Back substituting

$$\int u dv = uv - \int v du = (c^{n-1}) \left(-\frac{1}{2} e^{-c^2}\right) \Big|_0^\infty - \int_0^\infty -\frac{1}{2} e^{-c^2} (n-1) c^{n-2} dc$$

$$I_n = -\frac{c^{n-1}}{2} e^{-c^2} \Big|_0^\infty + \frac{(n-1)}{2} \int_0^\infty c^{n-2} e^{-c^2} dc$$

Need to use l'Hopital's rule $(\infty) \cdot (0)$

Can prove to yourself that $\rightarrow 0$

$$I_n = \frac{n-1}{2} \int_0^\infty c^{n-2} e^{-c^2} dc = \frac{n-1}{2} I_{n-2}$$

↳ Recursive Solution

Now we can have some fun:

$$I_0 = \frac{\sqrt{\pi}}{2} \quad I_1 = \frac{1}{2}$$

$$I_2 = \frac{\sqrt{\pi}}{4} \quad I_3 = \frac{1}{2}$$

$$I_4 = \frac{3}{8} \sqrt{\pi} \quad I_5 = 1$$

$$I_6 = \frac{15}{16} \sqrt{\pi} \quad \dots$$

$$\text{Example: } I_2 = \frac{2-1}{2} I_{2-2} = \frac{1}{2} I_0$$

$$= \frac{1}{2} \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{4}$$

Note, we haven't solved for I_1 , however need to solve it to get I_3, I_5, I_7 , etc...
To solve for I_1 , let $u = c^2$ & integrate. Try it

END OF LECTURE 2

Now we can look at each directional component of velocity

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(c_1) \phi(c_2) \phi(c_3) dc_1 dc_2 dc_3 = 1$$

This is true since the probability for three independent events is the product of the three individual probabilities.